

Assignment 2 Report:

Domain Adaptation with Small Language Models — A Shakespeare-Aware RAG System

CSCI933 Machine Learning Algorithms and Applications

Group Name — MLGong

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Abstract—This report describes a Shakespeare-aware language system developed for CSCI933 Assignment 2. The system addresses the domain adaptation challenge posed by Shakespearean text—archaic vocabulary, syntactic inversion, and cultural opacity—by combining dense embedding-based retrieval with a Small Language Model (SLM) generative component.

The implemented pipeline uses `sentence-transformers/all-MiniLM-L6-v2` to index 73 scene-level chunks drawn from *Hamlet*, *Macbeth*, and *Romeo and Juliet*. Retrieval is performed by cosine similarity over 384-dimensional dense vectors. Generation is handled by `google/flan-t5-large` (780 M parameters), conditioned on the top-3 retrieved passages. A TF-IDF keyword retrieval baseline provides a lower bound for comparison.

Structured evaluation on 11 questions (6 instructor-provided, 5 group-designed) using a 1–5 scale across correctness, grounding, retrieval relevance, and usefulness shows the RAG system outperforms the baseline on all criteria, with mean correctness improving from 1.9 to 3.9 and mean grounding improving from 1.0 to 3.6. Failure analysis identifies incomplete retrieval for character-specific questions, over-generalised generation, and limited Elizabethan style fidelity as the principal weaknesses.

I. INTRODUCTION AND PROBLEM FORMULATION

A. Problem Statement

The objective of this project is to build a Shakespeare-aware language system that enables a user with no prior exposure to Shakespeare to ask questions about three plays—*Hamlet*, *Macbeth*, and *Romeo and Juliet*—and receive accurate, grounded, beginner-friendly answers.

Shakespearean text presents a clear domain adaptation challenge: the lexicon is substantially Early Modern English (circa 1590–1610), and pre-trained language models trained on contemporary corpora have limited intrinsic competence on Shakespeare-specific factual questions without domain-specific augmentation.

B. Domain Adaptation as the Core Challenge

A pre-trained SLM presented with “Why does Hamlet delay?” may produce a plausible-sounding but inaccurate or non-specific answer, because (a) Shakespearean play structure and character motivations are under-represented in generic pre-training data, and (b) the model cannot distinguish Hamlet from

other literary princes without context. The principal adaptation mechanism in this work is *retrieval-augmented generation* (RAG): at inference time, the model is provided with relevant passages from the source text, so its generation is conditioned on actual textual evidence rather than prior associations.

C. System Objectives

The system is required to:

- answer concept-explanation questions (e.g. “Who is Ophelia?”);
- answer contextual questions requiring causal reasoning (e.g. “Why does Macbeth kill Duncan?”);
- display retrieved source passages alongside answers;
- generate short stylised responses in Elizabethan English when explicitly requested, clearly labelled as creative output.

D. Design Constraints

The system must run on a standard laptop or free-tier cloud compute (Google Colab). All models are pre-trained; no fine-tuning from scratch is performed. The chosen SLM (`flan-t5-large`, 780 M parameters) fits comfortably within the RAM budget of a free Colab instance and completes inference within seconds per query.

II. DATASET AND PREPROCESSING

A. Dataset Overview

The instructor-provided structured dataset covers three Shakespeare plays obtained from Project Gutenberg [4]. Each play is available in three formats: (a) a nested JSON of scenes and utterances (`hamlet.json`), (b) pre-chunked scene-level JSONL (`hamlet_scene_chunks.jsonl`), (c) utterance-level JSONL (`hamlet_utterances.jsonl`).

Table I summarises the dataset.

B. Dataset Schema

The raw data and the processed schema used in the retrieval system are described below.

Raw scene-chunk schema (as provided):

TABLE I
DATASET STATISTICS PER PLAY.

Play	Scenes	Utterances	Notes
Hamlet	21	1,337	5 acts; includes ghost scenes
Macbeth	28	1,290	5 acts; witches, murders
Romeo & Juliet	24	986	5 acts; balcony scene, duel
Total	73	3,613	

```
{
  "scene_id"      : "macbeth_1_3",
  "play"         : "Macbeth",
  "act"          : 1,
  "scene"        : 3,
  "scene_summary": "The witches prophesy Macbeth
    will be king.",
  "keywords"     : ["witches", "prophecy", "
    Macbeth"],
  "text"         : "MACBETH. So foul and fair a
    day ..."
}
```

Processed retrieval chunk schema:

```
{
  "chunk_id"      : "macbeth_1_3",
  "play"         : "Macbeth",
  "act"          : 1,
  "scene"        : 3,
  "scene_summary": "The witches prophesy Macbeth
    will be king.",
  "keywords"     : ["witches", "prophecy", "Macbeth
    "],
  "embed_text"   : "<summary> + <first 500 chars of
    text>",
  "display_text" : "<first 1200 chars of text>"
}
```

C. Chunking Strategy

We use **scene-level chunking**: each scene becomes one retrieval unit. This was chosen because:

- Shakespeare’s dramatic unit of meaning is the scene—a self-contained action with coherent character interaction;
- scene chunks are large enough to provide context for the language model, yet bounded enough to avoid overwhelming the 512-token input limit;
- the provided data already contains scene-level summaries and keywords that enrich the embedding signal.

The trade-off is that character-specific questions (e.g. about Ophelia) may surface scenes where that character is one participant among many, diluting retrieval precision. Utterance-level chunking would offer finer granularity but risks losing narrative context.

Embed text construction. Each chunk’s embedding is derived from: `scene_summary` field only. Using summary-only provides clean semantic signal without lexical noise from raw dialogue, which was found to cause motivation queries to retrieve aftermath scenes rather than deliberation scenes.

III. METHODOLOGY

A. Baseline System

The baseline uses **TF-IDF keyword retrieval** with no neural generation.

- 1) All 73 scene texts are indexed using a bigram TF-IDF vectoriser (`TfidfVectorizer`, max 10,000 features, English stop-word removal).
- 2) At query time, the query is transformed to a TF-IDF vector and cosine similarity is computed against the index.
- 3) The top-3 scene summaries are concatenated to form the “answer”—no generative model is involved.

This baseline is a fair lower bound because it represents the simplest possible retrieval-only approach: keyword matching with structured summarisation. It accesses the same source texts as the RAG system but uses no semantic understanding and no generation.

B. RAG-Based System

The RAG pipeline has four stages:

Stage 1: Embedding and Indexing. All 73 chunks are encoded with `sentence-transformers/all-MiniLM-L6-v2` (22 M parameters, 384-dimensional output). Embeddings are computed once and cached as a NumPy `.npz` file; subsequent runs load from cache.

Stage 2: Retrieval. Query text is encoded with the same model. Cosine similarity is computed between the query vector and the 73 stored chunk vectors using `sklearn.metrics.pairwise.cosine_similarity`. The top-3 chunks by similarity are returned.

Stage 3: Prompt Construction. The system prompt (Appendix B), retrieved passages, and user question are assembled into a single prompt string. Each passage is truncated to 300 characters to respect the 512-token limit of `flan-t5-large`.

Stage 4: Generation. `google/flan-t5-large` generates an answer conditioned on the full prompt, using beam search with $n = 4$ beams and `no_repeat_ngram_size=3` to prevent repetition. Maximum output length is 180 new tokens.

C. Model Choice and Justification

We selected `google/flan-t5-large` for the following reasons:

- **Genuine SLM.** At 780 M parameters it is genuinely small—orders of magnitude smaller than GPT-3 (175 B) or Claude.
- **Instruction fine-tuning.** Flan-T5 was explicitly trained on instruction-following tasks [1], making it suitable for structured Q&A without additional fine-tuning.
- **Free inference.** Runs on a Colab free T4 GPU (16 GB VRAM) with `float16` quantisation, or on CPU with modest latency.
- **Alignment with SLM principles.** The model is deployable on-premises, requires no proprietary API, and has no ongoing per-token cost—consistent with the assignment’s framing of small, auditable, resource-constrained systems.

IV. SYSTEM DESIGN AND IMPLEMENTATION

A. System Pipeline

The end-to-end flow is:

- 1) **User query** is entered as a plain English question.
- 2) **Query embedding** via MiniLM-L6-v2 (22 M params).
- 3) **Retrieval** of top-3 chunks by cosine similarity over the 73-vector index.
- 4) **Prompt construction**: system instruction + truncated passages + question.
- 5) **Response generation** by Flan-T5-Large (beam search, max 180 tokens).
- 6) **Evidence display**: retrieved play, act, scene, scene summary, and similarity score shown alongside the answer.
- 7) **Evaluation output**: structured CSV of answers, passages, and scores.

B. Implementation Details

- **Language**: Python 3.10
- **Libraries**: sentence-transformers 3.x, transformers 4.x, torch 2.x, scikit-learn, pandas, matplotlib
- **Execution**: Google Colab (free tier, T4 GPU or CPU)
- **Embedding cache**: embeddings_cache.npz (recomputed only if missing)
- **Output**: evaluation_results.csv and evaluation_chart.png

The repository follows the assignment’s recommended structure. To run the system: open assignment2_shakespeare_rag.ipynb in Google Colab, upload the dataset files (or mount Google Drive), and select *Runtime* → *Run all*.

Limitations. The 512-token input limit of Flan-T5 forces aggressive context truncation. Long scene texts are truncated to 350 characters per passage, potentially losing relevant dialogue. A model with a larger context window (e.g. Mistral-7B) would reduce this limitation.

V. EVALUATION DESIGN

A. Evaluation Questions

We evaluated both systems on 11 questions: 6 instructor-provided and 5 group-designed.

The instructor questions cover: Q1–Q2 (Macbeth motivation and character change), Q3–Q4 (Hamlet delay and Ophelia’s role), Q5–Q6 (Romeo and Juliet conflict).

The 5 group-designed questions test additional aspects:

- **G1**: Ghost exposition in Hamlet (tests retrieval of specific plot revelation).
- **G2**: Witches’ role in Macbeth (tests symbolic/thematic retrieval).
- **G3**: Friar Lawrence’s contribution (tests multi-scene causal reasoning across Romeo and Juliet).
- **G4**: Lady Macbeth’s sleepwalking (tests retrieval of a specific, named scene).

- **G5**: Shakespearean-style stylised response from Hamlet (tests creative generation quality).

B. Scoring Criteria

All answers are scored on a 1–5 integer scale:

Score	Label	Meaning
1	Poor	Incorrect, irrelevant, or unhelpful
2	Below average	Partially correct but misleading or unsupported
3	Acceptable	Mostly correct; some inaccuracies or weak grounding
4	Good	Correct, grounded, useful with minor gaps
5	Excellent	Fully correct, well-grounded, beginner-friendly

Criteria applied:

- **Correctness**: factual accuracy with respect to the plays
- **Grounding**: answer is traceable to retrieved passages
- **Retrieval relevance**: retrieved passages match the question topic
- **Usefulness**: helpful to a beginner with no prior Shakespeare knowledge
- **Style quality**: (G5 only) plausibly Elizabethan tone without sacrificing clarity

C. Evaluation Table Format

Table II shows mean scores. The full table is in Appendix A.

TABLE II
MEAN EVALUATION SCORES (1–5 SCALE) ACROSS 11 QUESTIONS.

System	Correctness	Grounding	Retrieval	Usefulness
Baseline	1.9	1.0	2.3	1.9
RAG System	3.9	3.6	4.4	3.8
Δ	+2.0	+2.6	+2.1	+1.9

VI. RESULTS AND ANALYSIS

A. Baseline versus RAG Comparison

Table II demonstrates consistent improvement across all criteria. The most striking gain is in **grounding** (+2.6 points): the baseline produces answers from scene summary concatenation with no traceability to specific passages, while the RAG system explicitly conditions generation on retrieved excerpts.

RAG most clearly helped for:

- **Q1** (Why does Macbeth kill Duncan?): RAG retrieved Act 1 Scene 7, where Macbeth’s soliloquy articulates his ambition. The generated answer correctly cited ambition, prophecy, and Lady Macbeth’s pressure, scoring 4/5 on correctness.
- **G1** (The Ghost in Hamlet): The Ghost scene (Act 1 Scene 5) was the top-retrieved passage, giving the model direct access to the murder revelation.
- **G2** (Witches in Macbeth): The retriever correctly identified Act 1 Scene 3 (the prophecy scene); the generated answer scored 5/5 on correctness.

RAG did *not* fully help for:

- **Q6** (Family feud in Romeo and Juliet): The answer generalised to “families in conflict” without citing Mercutio’s

death, Tybalt’s challenge, or the Prince’s banishment order. Retrieval relevance was 4/5 but correctness was 3/5.

- **G5** (Stylised Hamlet response): Flan-T5-Large is not specifically trained for Early Modern English, so style quality was only 3/5.

B. Qualitative Examples

Example 1 (Q1, successful):

- *Question:* Why does Macbeth kill Duncan?
- *Top retrieved passage:* Macbeth, Act 1 Scene 7 (“If it were done when ’tis done...”)
- *RAG answer:* Macbeth kills Duncan because of his ambition to become king, encouraged by the witches’ prophecy and pushed by Lady Macbeth. He knows it is wrong but convinces himself the prize justifies the act.
- *Interpretation:* Answer is factually correct, grounded in the retrieved soliloquy, and accessible to a beginner.

Example 2 (Q4, partial success):

- *Question:* What is Ophelia’s role in the tragedy?
- *Top retrieved passage:* Hamlet Act 3 Scene 1 (Polonius and Ophelia before Hamlet’s “To be or not to be”)
- *RAG answer:* Ophelia is caught between her love for Hamlet and her father Polonius’s commands. Her suffering shows how the people around Hamlet are hurt by the revenge plot.
- *Interpretation:* Partially correct; her death and madness (Act 4) were not retrieved. Retrieval relevance 3/5.

C. Failure Analysis

Failure 1 — Incomplete Retrieval (Q4). Character-specific questions about Ophelia retrieved Hamlet-centric scenes rather than Ophelia’s Act 4 arc. Scene-level chunking averages over multiple characters, diluting character-specific signal. *Remedy:* supplementary utterance-level index for character names; or keyword pre-filtering before dense retrieval.

Failure 2 — Over-generalised Generation (Q6). The model returned valid but non-specific text about “family conflict”. Flan-T5-Large can generate plausible general statements even when passages support more specific evidence. *Remedy:* prompt revision to require citation of specific events; increase top-*k* from 3 to 5.

Failure 3 — Limited Style Quality (G5). Flan-T5-Large is not instruction-tuned for Early Modern English. Output mixes archaic and modern constructions. *Remedy:* few-shot prompting with genuine Shakespeare examples; post-processing with a style classifier; or substituting a model fine-tuned on Elizabethan corpora.

VII. RESPONSIBLE USE OF GENERATIVE AI

Generative AI tools were used during this project for debugging assistance, documentation drafting, and evaluation prompt design. The following summarises usage (full log in Appendix D):

- **ChatGPT (GPT-4)** was used to suggest an initial chunking strategy. The suggestion of utterance-level chunking was

considered but replaced with scene-level chunks after manual inspection showed utterances were too short for meaningful RAG context.

- **GitHub Copilot** was used in VS Code for boilerplate code completion (tokeniser calls, batch encoding). All generated code was reviewed, tested, and modified.
- **Claude (Anthropic)** assisted with prompt wording for the system instruction. The suggested prompt was modified to add the 150-word limit for stylised output.

All AI-generated text was verified against the specification and the evaluated system behaviour. No AI-generated material was submitted without understanding and verification.

VIII. CONCLUSION

This project demonstrated that a modest RAG pipeline—dense retrieval with a 22 M-parameter embedding model and a 780 M-parameter SLM for generation—substantially improves question-answering performance over a keyword-baseline on Shakespearean domain questions. The RAG system improved mean correctness from 1.9 to 3.9 and mean grounding from 1.0 to 3.6 across 11 evaluation questions.

The most important design choice was scene-level chunking enriched with pre-computed scene summaries, which balanced contextual coherence against retrieval specificity. The second most important choice was using Flan-T5-Large, a genuine SLM that satisfies the assignment’s resource-constraint requirement without sacrificing Q&A competency.

With more time, we would: (a) add a parallel utterance-level index for character-specific queries; (b) increase retrieved chunks from 3 to 5 and use a model with a larger context window (e.g. Mistral-7B at 4-bit quantisation); (c) implement few-shot prompting for stylised generation; (d) conduct a richer failure analysis with inter-rater agreement to improve scoring reliability.

REFERENCES

- [1] H. W. Chung *et al.*, “Scaling instruction-finetuned language models,” *J. Mach. Learn. Res.*, vol. 25, no. 70, pp. 1–53, 2024. [Online]
- [2] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. EMNLP 2019*, pp. 3982–3992. [Online]
- [3] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020. [Online]
- [4] Project Gutenberg, “The Complete Works of William Shakespeare,” [Online]. Available: <https://www.gutenberg.org/>

APPENDIX A FULL EVALUATION TABLE

APPENDIX B REPRESENTATIVE PROMPTS

System instruction (non-stylised):

Answer the question in 2–3 clear sentences suitable for a beginner who has never read Shakespeare. Use the context provided.
Context 1 (Macbeth, Act 1, Scene 7): Macbeth debates whether to kill

TABLE III
FULL EVALUATION RESULTS — RAG SYSTEM (1–5 SCALE).

ID	Question (abridged)	Expected Focus	Corr.	Grd.	Ret.	Use.	Comments
Q1	Why does Macbeth kill Duncan?	Ambition, prophecy, Lady M	4	4	4	4	Act 1 Sc 7 retrieved; answer grounded in soliloquy
Q2	How does Macbeth change after kingship?	Paranoia, violence, isolation	4	4	5	4	Banquet scene retrieved; good coverage
Q3	Why does Hamlet delay?	Uncertainty, moral hesitation	4	4	4	4	Ghost scene + prayer scene retrieved
Q4	Ophelia’s role in the tragedy	Caught between men; cost of revenge	4	3	4	4	Act 4 death scenes not retrieved
Q5	Juliet’s conflict after Tybalt’s death	Loyalty vs. love	4	4	5	4	Act 3 Sc 2 retrieved; strong grounding
Q6	How does the feud shape the tragedy	Violence, secrecy, miscommunication	3	3	4	3	Over-generalised; specific events absent
G1	The Ghost in Hamlet	Murder revelation; duty of revenge	4	4	5	4	Act 1 Sc 5 top-retrieved
G2	Role of the three witches	Prophecy, fate, manipulation	5	4	5	4	Act 1 Sc 3 top-retrieved; strong
G3	Friar Lawrence’s contribution	Secret marriage; potion plan; miscommunication	4	3	4	4	Marriage scenes retrieved; potion plan weak
G4	Lady Macbeth’s sleepwalking	Guilt, collapse, contrast with earlier	4	4	5	4	Act 5 Sc 1 retrieved; good
G5	Stylised Hamlet response	Grief + duty in Early Modern English	3	3	3	3	Style partially Elizabethan; mixes modern
Mean			3.9	3.6	4.4	3.8	

TABLE IV
FULL EVALUATION RESULTS — BASELINE SYSTEM (1–5 SCALE).

ID	Question (abridged)	Corr.	Grd.	Ret.	Use.	Comments
Q1	Why does Macbeth kill Duncan?	2	1	2	2	Summary concatenation lacks causal explanation
Q2	How does Macbeth change?	2	1	2	2	Keywords match but no coherent answer
Q3	Why does Hamlet delay?	2	1	2	2	Scene summaries too vague
Q4	Ophelia’s role	2	1	2	2	No character-specific retrieval
Q5	Juliet’s conflict	2	1	3	2	RJ scenes retrieved; answer still vague
Q6	Family feud	2	1	2	2	Only keyword match; no causal content
G1	The Ghost in Hamlet	2	1	3	2	Ghost keywords matched; no explanation
G2	Role of witches	2	1	2	2	Witch scenes retrieved; no thematic synthesis
G3	Friar Lawrence	2	1	2	2	No Friar-specific retrieval
G4	Sleepwalking scene	2	1	3	2	Sleep scene partially retrieved
G5	Stylised Hamlet	1	1	2	1	No style generation; summary only
Mean		1.9	1.0	2.3	1.9	

Duncan. MACBETH. If it were done when ’tis done,
then ’twere well
It were done quickly. If the assassination ...

Question: Why does Macbeth kill Duncan?

Answer:

System instruction (stylised generation):

Answer the following question with a SHORT
Shakespearean-style
creative response (max 150 words, use thou/thee/
dost).
Label your response as creative output, not
textual evidence.
Base your response on the retrieved passages.

Evaluation rubric prompt (used for internal review):

You are a Shakespeare scholar evaluating a student

system.
Score the following answer on:
- Correctness (1-5): factual accuracy re: the
plays
- Grounding (1-5): supported by the retrieved
evidence
- Usefulness (1-5): helpful to a Shakespeare
beginner
Provide a brief justification for each score.

APPENDIX C ADDITIONAL OUTPUT EXAMPLES

G2 (Witches in Macbeth) — RAG System:

Retrieved: Macbeth Act 1 Sc 3 (“The witches prophesy Macbeth will be king”); Act 1 Sc 1 (“Fair is foul, and foul is fair”).

Generated answer: “The three witches play a central role in *Macbeth*. They open the play by predicting that Macbeth will

become Thane of Cawdor and eventually King of Scotland. These prophecies awaken Macbeth's ambition and set the events of the tragedy in motion. The witches also represent the theme of fate versus free will: the audience is left wondering whether the witches foretold events that were always going to happen, or whether they caused them by tempting Macbeth. Their famous phrase 'Fair is foul, and foul is fair' captures the moral confusion that runs through the entire play."

Scores: Correctness 5, Grounding 4, Retrieval 5, Usefulness 4.

APPENDIX D GENAI USAGE LOG

TABLE V
GENAI USAGE LOG.

Tool	Prompt / Task	Useful Output	Verification / Modification
ChatGPT (GPT-4)	“What are the best chunking strategies for RAG on Shakespeare?”	Suggested utterance-level, scene-level, and overlapping fixed-size chunks; explained trade-offs.	Tested scene-level empirically; replaced utterance suggestion with scene-level after finding utterances too short (median 12 words).
GitHub Copilot	Boilerplate for tokeniser batch encoding and cosine similarity call	Inline code suggestion for <code>model.encode(texts, batch_size=32)</code>	Reviewed and tested; adjusted <code>batch_size</code> and removed redundant <code>.tolist()</code> call.
Claude (Anthropic)	“Write a system prompt for a Shakespeare RAG chatbot targeting beginners”	Suggested prompt with “Do not invent unsupported details” and “If context is insufficient, say so” clauses.	Added the 150-word cap for stylised output; added “label stylised output as creative, not evidence”.
ChatGPT (GPT-4)	“Review this evaluation rubric for a RAG Shakespeare system”	Suggested adding a ‘retrieval relevance’ criterion separate from grounding.	Incorporated retrieval relevance as a distinct scoring dimension; verified it matched assignment spec criteria.